

Question	Marking Guidance	Mark	Comments
08.1	1. One glycerol and three fatty acids; 2. Condensation (reactions) and removal of three molecules of water; 3. Ester bond(s) (formed);	3	Accept all marks in suitably labelled diagram OR in a balanced equation

Question	Marking Guidance	Mark	Comments
08.2	Palmitoleic acid is an unsaturated fatty acid represented by diagram K ;	1	

Question	Marking Guidance	Mark	Comments
08.3	1. To increase accuracy/resolution because differences/lengths are small; 2. To increase accuracy because reduces risk of human error; 3. To increase accuracy because roots are less (likely to be) damaged; 4. To reduce error/uncertainty because differences/lengths are small;	1 max	Ignore 'precision'

Question	Marking Guidance	Mark	Comments
08.4	1. Population 1 grew longer roots in warm temperatures and population 2 grew longer roots in cool temperatures; 2. Standard deviations do not overlap so difference (in mean) unlikely to be/not due to chance; 3. Population 1 (is better adapted to warm conditions because it) has more saturated fatty acids so more energy available (and more growth); 4. Population 2 (is better adapted to cool conditions because it) has more unsaturated/liquid fatty acids so more lipase activity (and more growth);	4	2. Accept: 'Standard deviations do not overlap showing difference (in mean likely to be) significant' 3. and 4. Accept for 'fatty acids', fat

Question	Marking Guidance	Mark	Comments
08.5	Same species OR (If mated) can produce fertile offspring OR (It is) genus and species name;	1	